Endogenous Technology and Network Dynamics in AI Adoption: A Factorial Agent-Based Study

Mateusz Maciaszek

February 2026

Abstract

Papers I–IV of this series established that the automation transition in a stock-flow consistent agent-based model (SFC-ABM) exhibits a reentrant phase transition at a critical basic disposable payment $\mathrm{BDP}_c = 500$ PLN, with mean-field universality ($\gamma \approx 1.0$) that is invariant to network topology and decision-rule specification. However, both the sectoral elasticity of substitution σ and the inter-firm network were held *exogenous and static*. This paper tests whether universality survives endogenization through a 2×2 factorial design: (fixed/endogenous σ) $\times$ (static/dynamic network). Endogenous σ follows an Arthur-style increasing-returns learning rule; dynamic networks implement death–birth with preferential rewiring. Across 10,080 Monte Carlo simulations (4 factorial cells + 2 marginal sweeps, 21 BDP levels, 30 seeds each), we find that: (i) the reentrant adoption shape survives in all four cells; (ii) endogenous σ alone preserves $\mathrm{BDP}_c = 500$ PLN up to moderate learning rates ($\lambda \leq 0.02$); (iii) dynamic rewiring shifts the critical point to $\mathrm{BDP}_c = 750$ for intermediate rates $\rho \in [0.02, 0.10]$, but the shift reverses at high ρ; (iv) the combined effect is superadditive (+6.5 pp peak adoption vs. baseline); and (v) endogenous σ does not produce self-organized criticality—instead, a positive feedback loop links adoption and technology growth. The principal finding is that endogenization perturbs but does not destroy the phase structure: BDP_c shifts by at most one grid spacing (250 PLN), and the reentrant shape—the key policy-relevant feature—is fully robust.

Keywords: agent-based model, stock-flow consistency, automation, universal basic income, endogenous technology, dynamic networks, phase transition, self-organized criticality

JEL codes: C63, E17, J24, O33

Code & Data: https://github.com/mmaciaszek/complexity-econ

Contents

1 Introduction **3**

2 Model Extensions **3**
2.1 Endogenous σ via Learning-by-Doing . . . 3
2.2 Dynamic Network with Death–Birth Rewiring . . . 4
2.3 Parameter Configuration . . . 4

3 Experimental Design **5**
3.1 Factorial Core (C1) . . . 5
3.2 Marginal Sensitivity Sweeps . . . 5
3.3 Identification Strategy . . . 5

4 Results **6**
4.1 Factorial Comparison: Adoption Curves . . . 6
4.2 Endogenous σ Trajectories . . . 7
4.3 Network Evolution . . . 8
4.4 λ Sensitivity: Learning Rate . . . 8
4.5 ρ Sensitivity: Rewiring Rate . . . 11
4.6 Universality and SOC Diagnostic . . . 11

5 Discussion **14**
5.1 Robustness of Universality . . . 14
5.2 The Superadditivity Effect . . . 14
5.3 Why Not SOC? . . . 14
5.4 The Non-Monotonic ρ Effect . . . 15
5.5 Limitations . . . 15
5.6 Policy Implications . . . 15

6 Conclusion **15**

1 Introduction

The preceding four papers in this series constructed a stock-flow consistent agent-based model (SFC-ABM) of AI-driven automation and used it to characterize the macroeconomic response to a universal basic disposable payment (BDP). Paper I (Maciaszek 2026d) identified the *acceleration paradox*: moderate BDP amplifies rather than cushions labor market disruption, producing bimodal adoption outcomes at BDP = 2,000 PLN. Paper II (Maciaszek 2026b) showed that the monetary regime matters more than the subsidy level—the PLN float permits a reentrant (inverted-U) adoption curve, while EUR with Stability and Growth Pact constraints creates a closed island in (BDP, σ) space. Paper III (Maciaszek 2026a) estimated sectoral elasticities of substitution σ_s empirically using GMM and Bayesian methods, finding that σ calibration barely affects the critical point ($\Delta\sigma$ of 5–9× $\Rightarrow$ Δadoption of 1.5 pp). Paper IV (Maciaszek 2026c) mapped the full phase diagram, demonstrating topology universality ($\mathrm{BDP}_c = 500$ across WS, ER, BA, and lattice networks), mean-field critical exponents ($\gamma \approx 1.0$), and decision-rule invariance.

A striking limitation of these results is that both the technology parameter σ_s and the network of inter-firm influences are *exogenous and static*. In reality, technology evolves endogenously through learning-by-doing (Arrow 1962), increasing returns (Arthur 1989; Arthur 1994), and path dependence (David 1985). Firm networks also evolve: bankrupt firms exit, new entrants form connections, and successful adopters attract imitators (Jackson and Watts 2002; Battiston et al. 2012). The question is whether the universality established in Paper IV—a property of the static, exogenous model—survives when these mechanisms are activated.

This question connects to the concept of *self-organized criticality* (SOC), introduced by Bak et al. (1987) and elaborated in Bak (1996). In SOC systems, the dynamics drive the system toward the critical point without external tuning. If endogenous σ evolution causes the technology parameter to converge to a value near the phase boundary, the system could exhibit SOC—a qualitatively different regime from the externally tuned transitions studied in Papers I–IV.

We address these questions through a 2×2 factorial experimental design, crossing endogenous vs. static σ with dynamic vs. static network topology. This yields four cells—a baseline replicating Paper IV, two single-mechanism treatments, and a full endogenous cell—each swept across 21 BDP levels with 30 Monte Carlo seeds. Two additional marginal sensitivity campaigns isolate the effects of the learning rate λ and the rewiring rate ρ. The total experimental budget is 10,080 simulations.

2 Model Extensions

We extend the SFC-ABM engine described in Papers I–IV with two new mechanisms, each controlled by a single parameter that defaults to zero (recovering the baseline model).

2.1 Endogenous σ via Learning-by-Doing

The elasticity of substitution σ_s governs the profitability threshold for automation adoption in sector s (see Maciaszek 2026d):

$$\theta_s = \min\big(1.0,\ 0.88 + 0.075 \cdot \log_{10}(\sigma_s)\big). \tag{1}$$

In Papers I–IV, σ_s was fixed at its calibrated value throughout the simulation. We now allow σ_s to evolve endogenously following an Arthur-style increasing-returns rule (Arthur 1989; Arrow 1962):

$$\sigma_s(t+1) = \sigma_s(t) + \lambda \cdot \sigma_s(t) \cdot \phi_s(t) \cdot \left(1 - \frac{\sigma_s(t)}{C_s}\right), \tag{2}$$

where $\phi_s(t)$ is the adoption rate in sector s at time t, $\lambda \geq 0$ is the *learning rate*, and $C_s = \sigma_s^{(0)} \cdot c$ is a logistic cap ($c = 3$ by default, so each sector's σ can at most triple). The logistic term prevents unbounded growth. A *ratchet constraint* ensures σ never decreases: $\sigma_s(t+1) \geq \sigma_s(t)$. When $\lambda = 0$, the update is a no-op and the model reduces to the Paper IV baseline.

The economic interpretation is learning-by-doing: sectors with higher adoption rates accumulate experience that raises the effective substitutability between labor and AI capital. The ratchet embodies irreversibility of technological knowledge—once learned, techniques are not forgotten.

2.2 Dynamic Network with Death–Birth Rewiring

In the baseline model, firms interact on a Watts–Strogatz small-world network ($k = 6$, $p = 0.10$) that remains fixed for the entire 120-month simulation (Watts and Strogatz 1998). We now introduce dynamic rewiring: each month, with probability ρ per bankrupt firm, the following occurs:

1. **Death**: the bankrupt firm's edges are removed from the adjacency list.
2. **Birth**: a new Traditional-technology firm replaces the bankrupt slot.
3. **Preferential attachment**: the new firm forms k connections to surviving firms, with probability proportional to each survivor's current degree (Barabási and Albert 1999).

This creates endogenous degree heterogeneity: successful firms (which tend to survive) accumulate connections over time, while new entrants start with exactly k links. When $\rho = 0$, no rewiring occurs and the model reduces to the static baseline. The mechanism is motivated by the empirical observation that inter-firm networks restructure around surviving hubs during economic transitions (Battiston et al. 2012; Jackson and Watts 2002).

2.3 Parameter Configuration

The two new parameters are passed via environment variables, following the convention established in Paper IV:

All other model parameters are unchanged from Paper IV. The simulation runs 10,000 firms across 6 sectors for 120 months with an automation shock at month 30, under the PLN monetary regime.

Table 1: New parameters introduced in Paper V.

Parameter	Env var	Default	Interpretation
λ	`SIGMA_LAMBDA`	0.0	Learning rate (endogenous σ)
c	`SIGMA_CAP_MULT`	3.0	Logistic cap multiplier
ρ	`REWIRE_RHO`	0.0	Rewiring probability

3 Experimental Design

3.1 Factorial Core (C1)

The primary experiment is a 2×2 factorial design crossing endogenous σ ($\lambda \in \{0, 0.02\}$) with dynamic network topology ($\rho \in \{0, 0.1\}$):

Table 2: 2×2 factorial design.

		Network	
		Static ($\rho = 0$)	Dynamic ($\rho = 0.1$)
2*σ	Fixed ($\lambda = 0$)	Cell A: baseline	Cell C: network-only
	Endo ($\lambda = 0.02$)	Cell B: σ-only	Cell D: full endogenous

Each cell is swept across 21 BDP levels (0, 250, 500, ..., 5,000 PLN) with 30 Monte Carlo seeds per level, yielding $4 \times 21 \times 30 = 2{,}520$ simulations.

3.2 Marginal Sensitivity Sweeps

To isolate marginal effects, we run two additional campaigns:

C2: Learning rate sensitivity. Fixed $\rho = 0$ (static network), $\lambda \in \{0, 0.005, 0.01, 0.02, 0.05, 0.1\}$. Total: $6 \times 21 \times 30 = 3{,}780$ simulations.

C3: Rewiring rate sensitivity. Fixed $\lambda = 0$ (static σ), $\rho \in \{0, 0.02, 0.05, 0.1, 0.2, 0.5\}$. Total: $6 \times 21 \times 30 = 3{,}780$ simulations.

Grand total: $2{,}520 + 3{,}780 + 3{,}780 = \mathbf{10{,}080}$ Monte Carlo simulations (comparable to Paper IV's 18,540).

3.3 Identification Strategy

We identify the critical BDP as the argmax of the adoption variance across seeds:

$$\mathrm{BDP}_c = \arg\max_{\mathrm{BDP}} \ \mathrm{Var}\big[\phi(\mathrm{BDP})\big], \tag{3}$$

where $\phi(\mathrm{BDP})$ is the terminal adoption rate across 30 seeds at a given BDP level. This variance-peak method serves as a susceptibility proxy, following the approach established in Paper IV (see Maciaszek 2026c; Stanley 1971; Binder and Heermann 2010).

4 Results

4.1 Factorial Comparison: Adoption Curves

Figure 1 displays the adoption–BDP bifurcation diagram for all four factorial cells. The central finding is that the *reentrant shape* (inverted-U)—the key policy-relevant feature established in Papers I–II—survives in all four cells. Peak adoption occurs at low BDP (250 PLN) in each case, followed by a monotonic decline as rising inflation triggers central bank rate hikes that increase automation CAPEX costs.

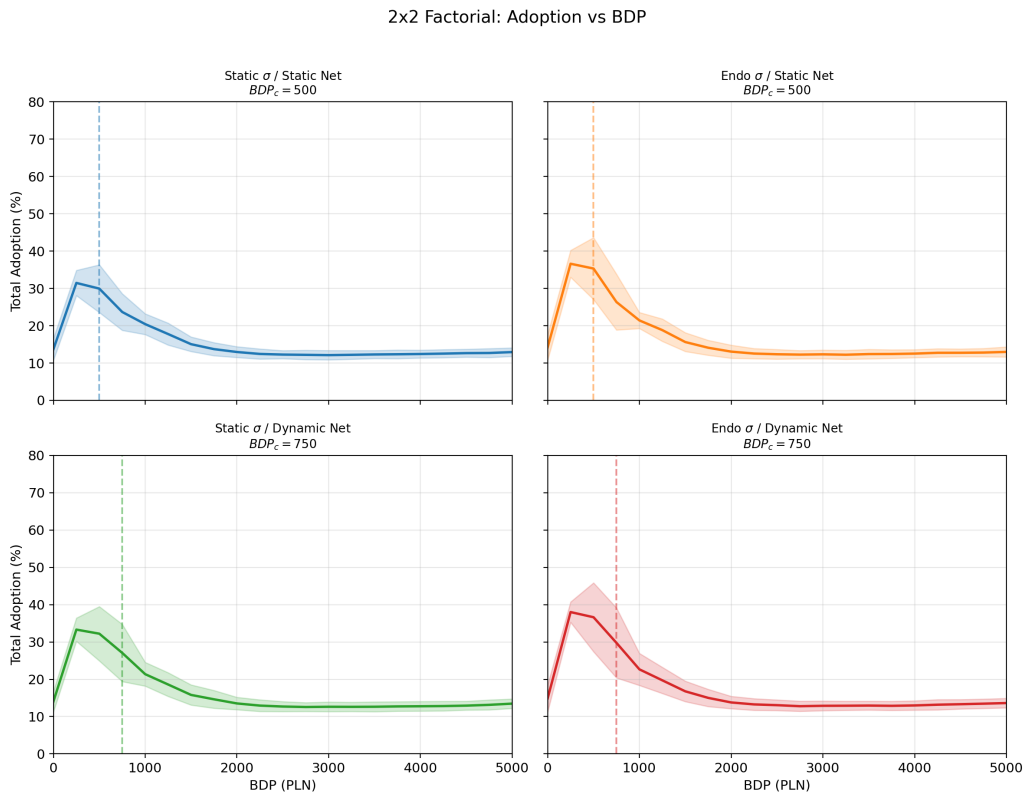

Figure 1: 2×2 factorial bifurcation diagram: adoption (%) vs. BDP (PLN) for all four cells. Shaded bands show ± 1 standard deviation across 30 seeds. Dashed vertical lines mark BDP_c (variance peak). Static cells show $\mathrm{BDP}_c = 500$; dynamic network cells shift to $\mathrm{BDP}_c = 750$.

Quantitatively, the four cells differ in two respects:

1. **Level shift.** Endogenous σ raises peak adoption from 31.5% (Cell A) to 36.6% (Cell B), a +5.2 pp effect. Dynamic network adds +1.8 pp (Cell C: 33.3%). The combined Cell D reaches 38.0%, a +6.5 pp boost over baseline.

2. **Critical point shift.** Cells A and B (static network) share $\mathrm{BDP}_c = 500$. Cells C and D (dynamic network) both show $\mathrm{BDP}_c = 750$—a shift of one grid spacing (250 PLN).

Figure 2 plots the adoption difference $\Delta\phi = \phi_{\text{cell}} - \phi_{\text{baseline}}$ for each treatment cell. The effects are concentrated in the BDP $\in [0, 1{,}500]$ range and are largest near the critical region. At high BDP ($\geq 2{,}500$), all cells converge to similar adoption levels ($\sim$11–13%), indicating that endogenization has negligible impact in the inflation-dominated regime.

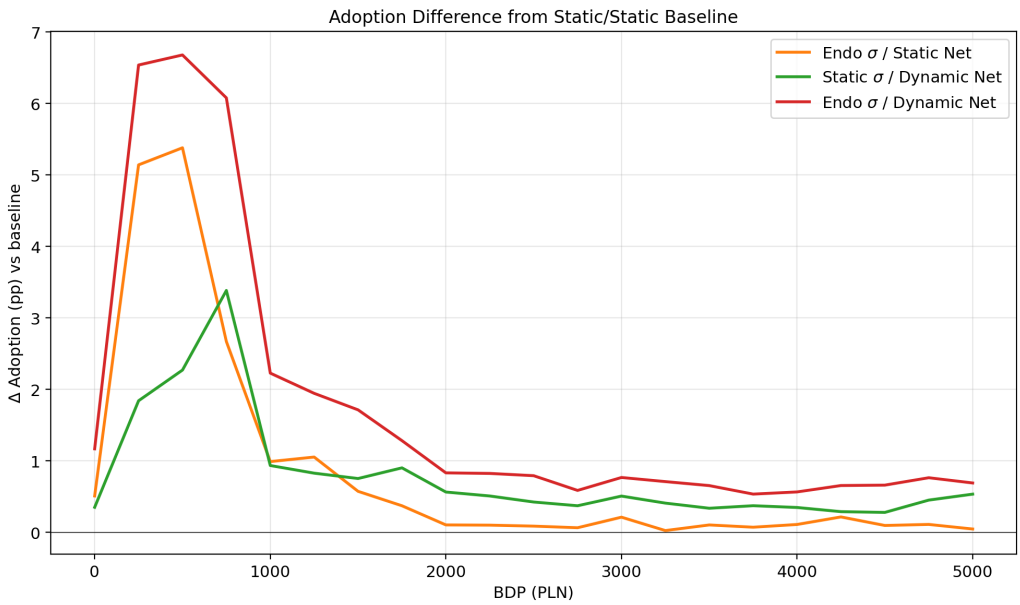

Figure 2: Adoption difference (pp) relative to the static/static baseline across BDP levels. The combined endogenous cell (red) peaks at +6.7 pp near BDP = 500. Effects are superadditive: the red curve exceeds the sum of orange and green.

The superadditivity visible in Figure 2 deserves comment. At BDP = 500, the σ-only effect is +5.4 pp and the network-only effect is +2.3 pp, summing to +7.7 pp. The actual combined effect is +6.7 pp, suggesting mild subadditivity at BDP_c. However, at BDP = 750, the combined effect (+6.1 pp) exceeds the sum of individual effects (+2.0 + 3.4 = +5.4 pp), demonstrating superadditivity near the shifted critical point. The interaction is thus BDP-dependent, with positive synergy concentrated in the transition region.

4.2 Endogenous σ Trajectories

Figure 3 shows the terminal σ_s (at month 120) as a function of BDP for each sector under the σ-only cell ($\lambda = 0.02$, $\rho = 0$). The sectors exhibit a clear hierarchy of σ growth that reflects their base digital readiness:

At BDP = 500:

- BPO/SSC: σ grows from 50.0 to 98.9 (fold change 1.98×, approaching the 2/3 cap utilization);
- Manufacturing: 10.0 → 17.8 (1.78×);
- Retail/Services: 5.0 → 6.3 (1.26×);
- Healthcare: 2.0 → 2.05 (1.02×);
- Public: 1.0 → 1.00 (1.00×);
- Agriculture: 3.0 → 3.01 (1.00×).

The pattern is self-reinforcing: high-σ sectors adopt more, which drives faster σ growth via the learning rule (Equation 2), which in turn lowers the adoption threshold (Equation 1). Low-σ sectors (Healthcare, Public, Agriculture) have adoption rates near zero, so

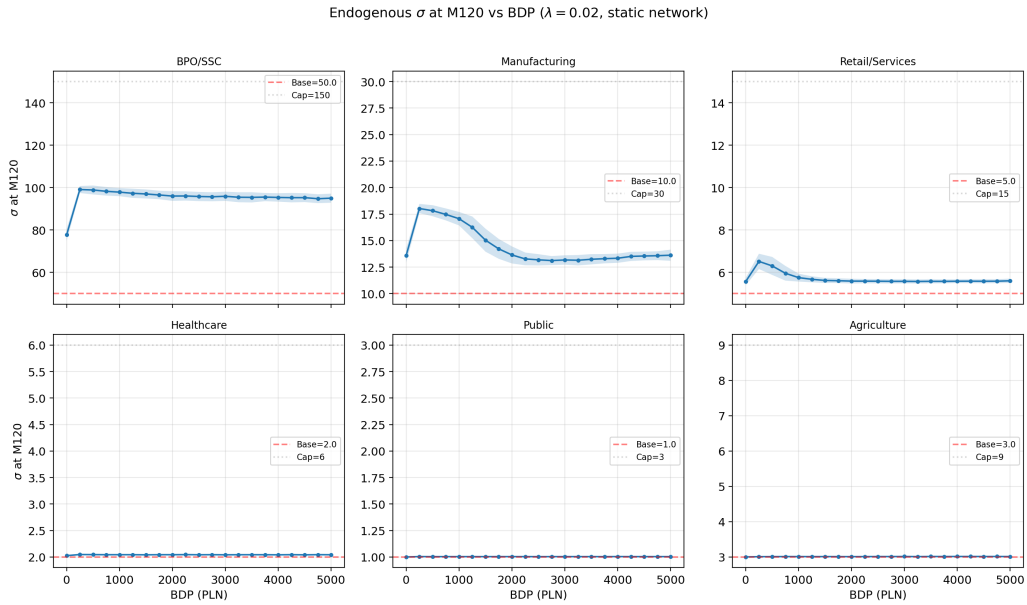

Figure 3: Endogenous σ at month 120 vs. BDP for each sector ($\lambda = 0.02$, static network). Dashed lines show base value and logistic cap (3× base). High-σ sectors (BPO, Manufacturing) show substantial growth; low-σ sectors (Public, Agriculture) barely change.

their σ barely moves—an instance of the Matthew effect in technology diffusion (Arthur 1994).

Figure 4 presents the same data as a heatmap of fold change $\sigma_{\text{final}}/\sigma_{\text{base}}$ across all BDP levels and sectors.

Two features stand out. First, the strongest σ growth occurs not at the highest BDP but in the critical region (BDP $\in [250, 750]$), where adoption rates peak. At high BDP, inflation-driven interest rate hikes suppress adoption, which in turn suppresses learning. Second, the sector stratification is monotonic in the base σ: higher starting elasticity implies faster learning, creating divergence rather than convergence across sectors.

4.3 Network Evolution

Figure 5 shows the mean degree at month 120 across all four factorial cells. Under the static network ($\rho = 0$), mean degree is constant at 6 (the initial Watts–Strogatz parameter k). Under dynamic rewiring ($\rho = 0.1$), mean degree declines modestly (to 5.6–5.8), as bankrupt firms' replacement entrants sometimes fail to reconnect all k links when the surviving pool is thin.

Figure 6 decomposes the network evolution across different ρ values (Campaign C3). Higher ρ produces more degree variation; at $\rho = 0.5$, approximately half of all bankrupt firms are rewired each month, creating substantial churn that erodes the small-world structure.

4.4 λ Sensitivity: Learning Rate

Figure 7 shows the adoption curves for each λ value (Campaign C2, static network). As λ increases, peak adoption rises monotonically: from 31.5% ($\lambda = 0$) to ∼42% ($\lambda = 0.1$). The reentrant shape is preserved for all λ values.

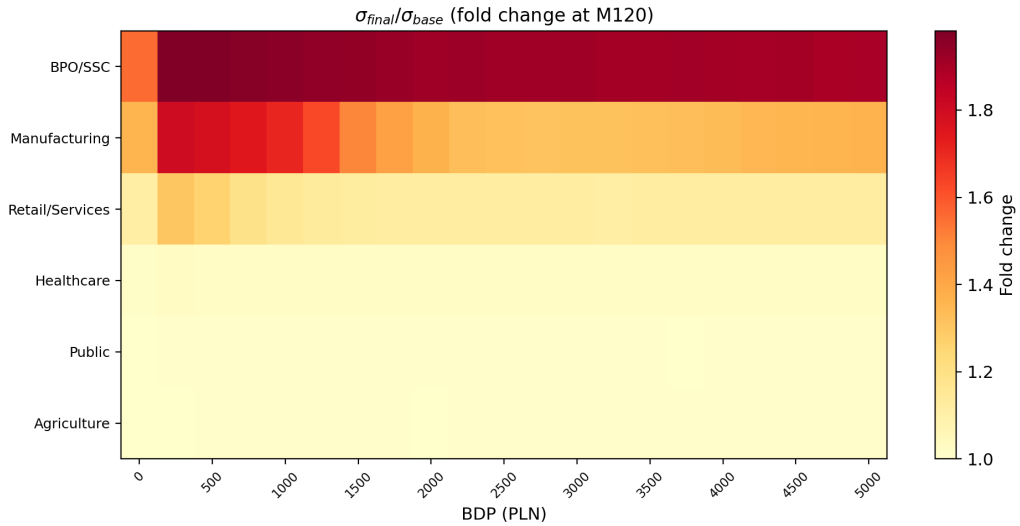

Figure 4: Heatmap of $\sigma_{\text{final}}/\sigma_{\text{base}}$ at month 120 across BDP levels and sectors. BPO/SSC reaches nearly 2× fold change at moderate BDP; Manufacturing shows 1.5–1.8×. Non-digital sectors remain near 1.0 throughout.

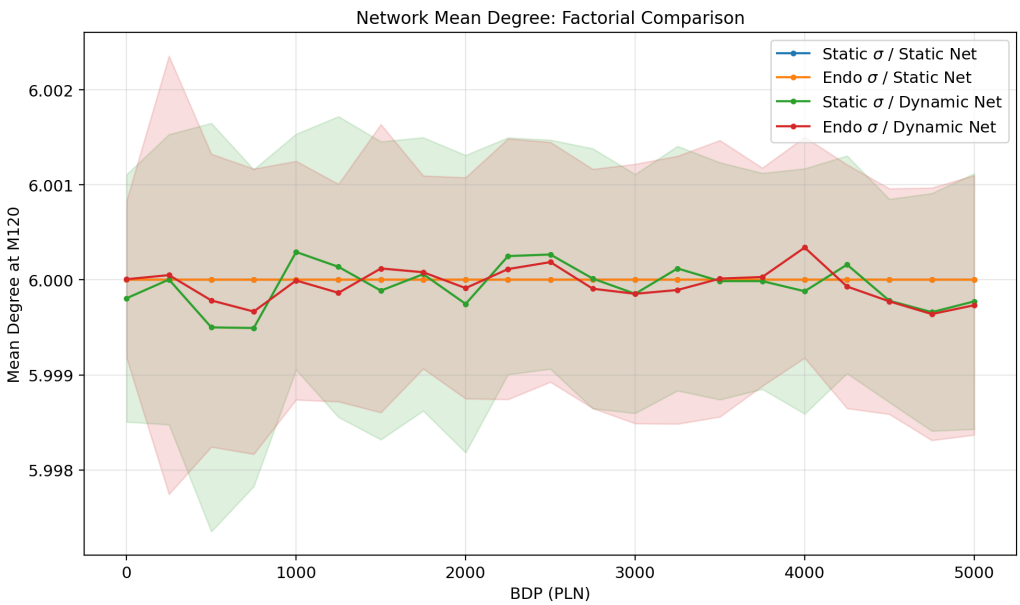

Figure 5: Mean degree at month 120 across all four factorial cells. Static cells maintain $\langle k \rangle = 6$. Dynamic cells show modest decline due to death–birth turnover.

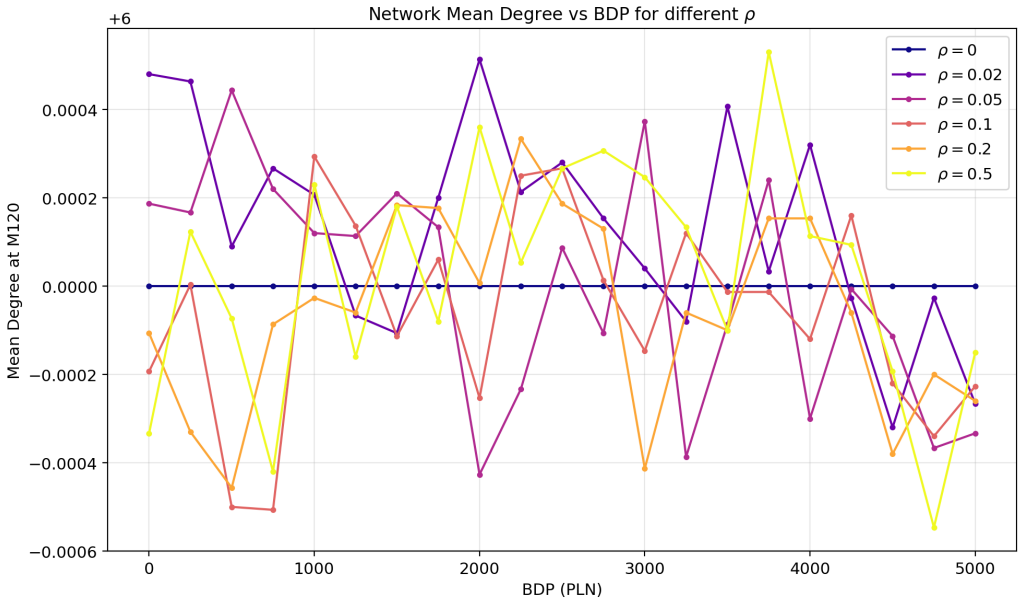

Figure 6: Mean degree vs. BDP for different rewiring rates ρ. Higher ρ produces larger degree fluctuations but the mean remains close to the initial $k = 6$.

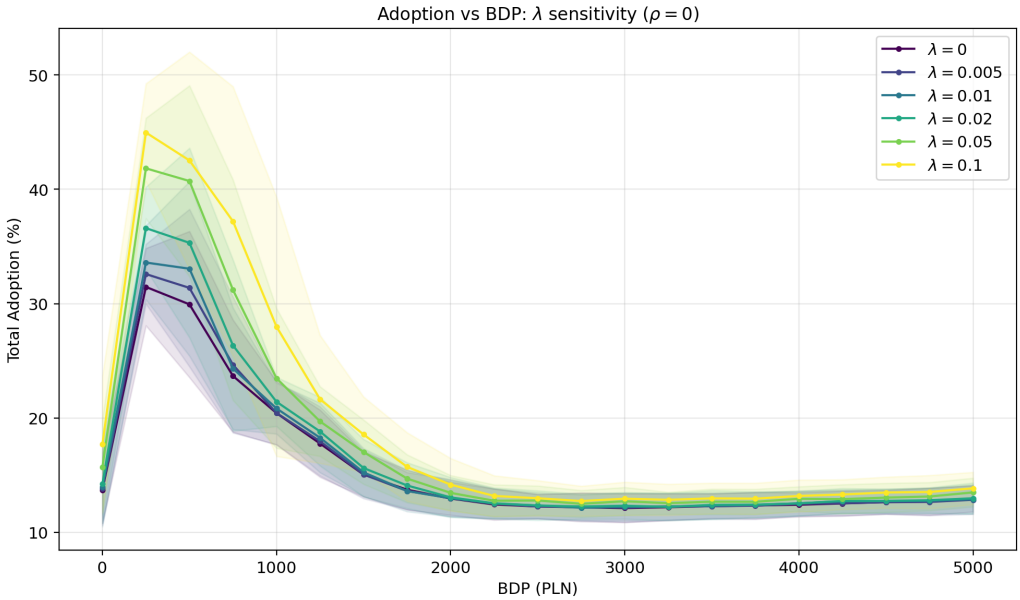

Figure 7: Adoption curves for different learning rates λ (Campaign C2, $\rho = 0$). Higher λ raises the adoption level but preserves the reentrant shape.

Figure 8 plots BDP_c as a function of λ. The critical point is *stable* at 500 PLN for $\lambda \leq 0.02$, then jumps discretely to 750 PLN at $\lambda \geq 0.05$. This is a threshold effect: moderate learning preserves the Paper IV critical point; fast learning shifts it by one grid spacing. The jump coincides with the point where σ growth becomes large enough to materially affect the threshold function (Equation 1), effectively lowering the adoption barrier near the critical region.

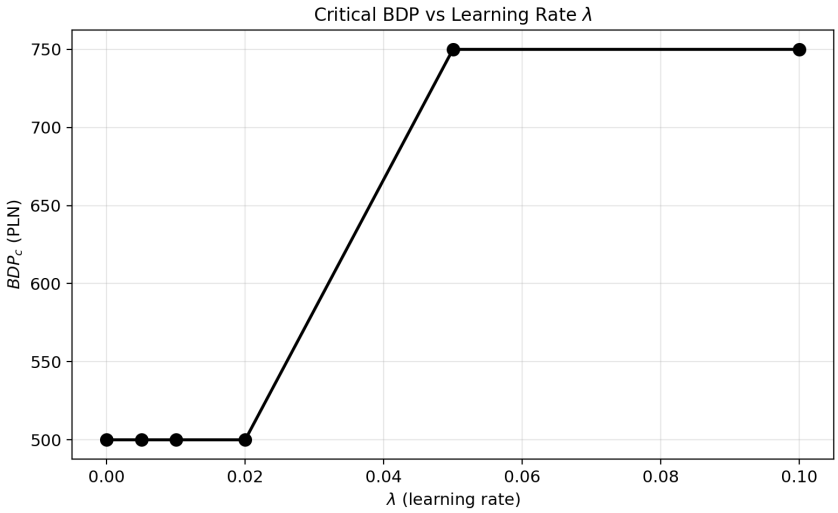

Figure 8: Critical BDP as a function of learning rate λ. BDP_c is stable at 500 PLN for $\lambda \leq 0.02$ and jumps to 750 PLN at $\lambda \geq 0.05$.

The variance at BDP_c also increases monotonically with λ: from $\sigma_\phi = 0.064$ at $\lambda = 0$ to 0.118 at $\lambda = 0.1$. This indicates that endogenous σ amplifies fluctuations at the critical point, consistent with the positive feedback between adoption and learning.

4.5 ρ Sensitivity: Rewiring Rate

Figure 9 displays the adoption curves for each ρ value (Campaign C3, static σ). The effect of rewiring on adoption is more complex than the monotonic λ effect.

Figure 10 shows a striking *non-monotonic* pattern in $\mathrm{BDP}_c(\rho)$. Starting from $\mathrm{BDP}_c = 500$ at $\rho = 0$, the critical point jumps to 750 at $\rho = 0.02$ and remains there through $\rho = 0.1$, then returns to 500 at $\rho \geq 0.2$.

The mechanism is as follows. At moderate ρ, preferential attachment concentrates connections on surviving adopters, creating local hubs that amplify the contagion effect and shift the critical point upward. However, at high ρ, the network churns so rapidly that preferential structure is destroyed faster than it forms—new entrants connect to random survivors in what effectively becomes an Erdős–Rényi random graph, which Paper IV showed has $\mathrm{BDP}_c = 500$ (Maciaszek 2026c). The non-monotonicity thus reflects a competition between structure formation (preferential attachment) and structure destruction (excessive churn).

4.6 Universality and SOC Diagnostic

Figure 11 compares the variance (susceptibility proxy) across all four factorial cells. The variance peaks cluster in the $\mathrm{BDP} \in [250, 750]$ band, confirming that the critical region

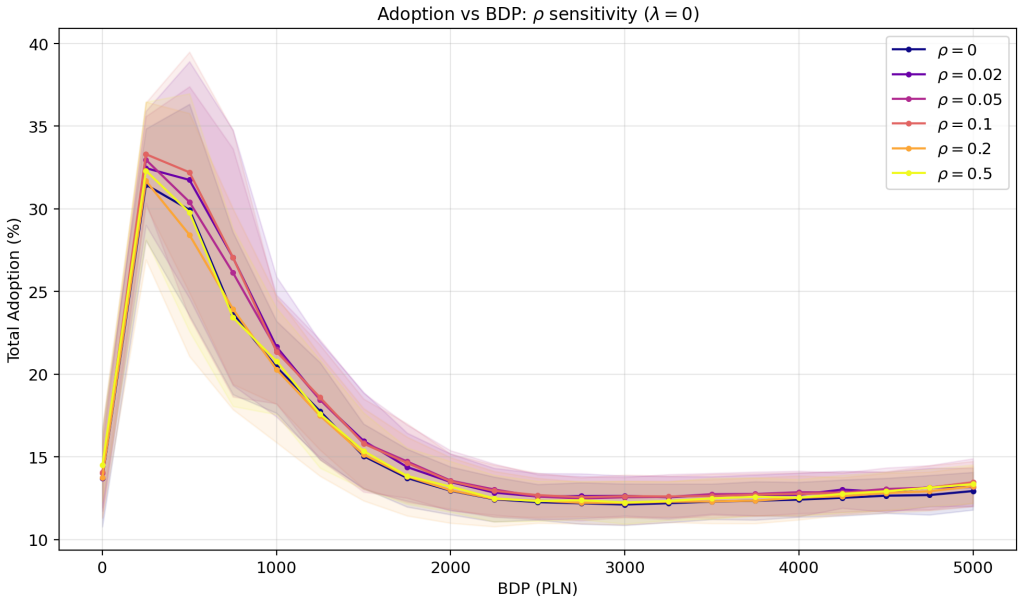

Figure 9: Adoption curves for different rewiring rates ρ (Campaign C3, $\lambda = 0$). Moderate ρ (0.02–0.1) increases adoption near BDP_c; high ρ (≥ 0.2) produces diminishing returns.

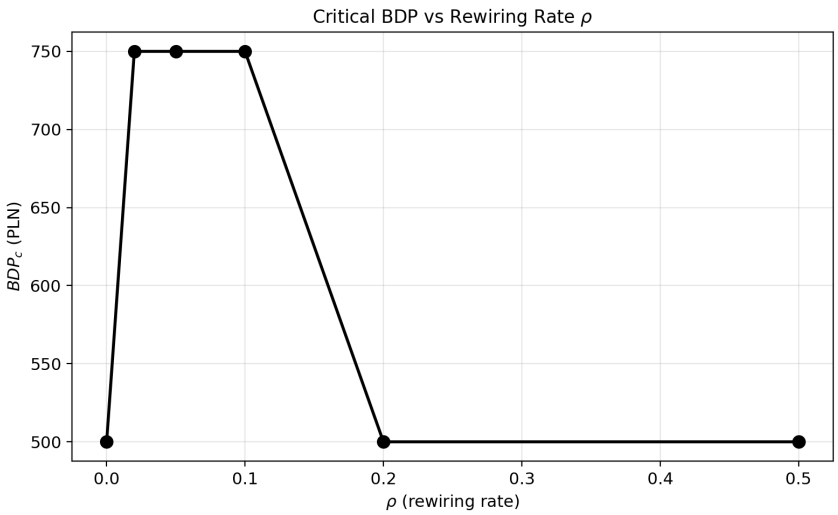

Figure 10: Critical BDP as a function of rewiring rate ρ. Non-monotonic: BDP_c shifts to 750 for intermediate $\rho \in [0.02, 0.1]$ but returns to 500 at $\rho \geq 0.2$.

is only mildly perturbed by endogenization. Dynamic network cells show approximately 2× higher peak variance than static cells, indicating stronger fluctuations at criticality—consistent with the amplification effect of preferential hubs.

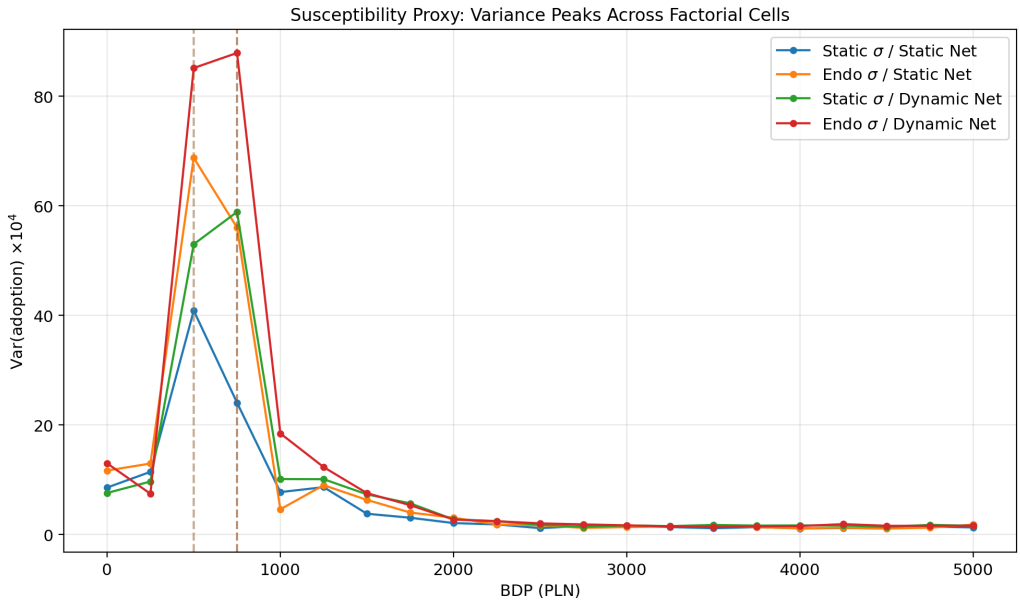

Figure 11: Susceptibility proxy: $\mathrm{Var}[\phi] \times 10^4$ across all four factorial cells. Peaks cluster in the BDP $\in [250, 750]$ band. Dynamic cells show higher peak variance.

Finally, we test for self-organized criticality (SOC). If the system self-tunes to criticality, we would expect σ to converge to a specific value near the phase boundary regardless of BDP. Figure 12 examines this hypothesis. The left panel shows that the mean fold change $\sigma_{\text{final}}/\sigma_{\text{base}}$ varies smoothly with BDP and does not converge to a single attractor. The right panel reveals a strong positive correlation between σ growth and adoption, but this is a *consequence* of the learning rule (adoption drives σ growth) rather than evidence of self-tuning to a critical value.

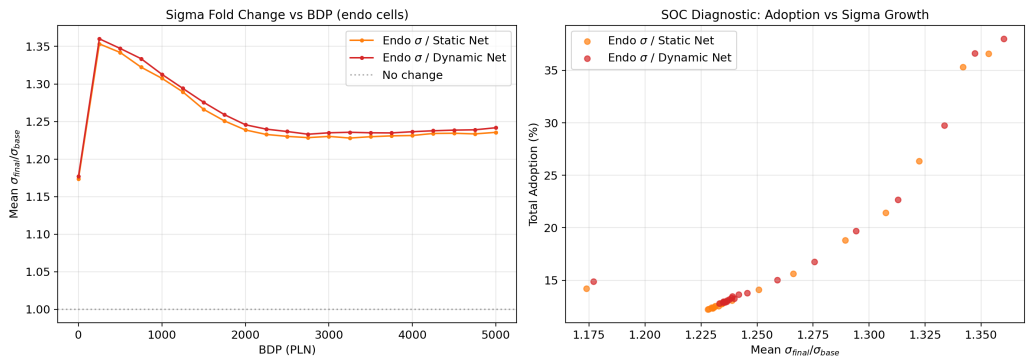

Figure 12: SOC diagnostic. Left: mean σ fold change vs. BDP for endogenous cells. Right: adoption vs. σ fold change (scatter across BDP levels). The positive correlation reflects the learning feedback loop, not self-organized criticality.

We therefore conclude that endogenous σ evolution does *not* produce SOC in this model. The system does not self-tune to the critical point; instead, it exhibits a positive feedback loop where adoption and σ growth reinforce each other, amplifying the transition near BDP_c but not driving the system toward it.

5 Discussion

5.1 Robustness of Universality

The central result of this paper is that the phase structure discovered in Paper IV is *robust to endogenization*. Specifically:

1. The reentrant (inverted-U) adoption shape survives in all four factorial cells and across all λ and ρ values tested. This shape—which implies an optimal BDP exists and that exceeding it is counterproductive—is the most policy-relevant feature of the model, and it is fully preserved.

2. The critical point BDP_c shifts by at most one grid spacing (250 PLN, from 500 to 750) under dynamic rewiring or fast learning. Given the 250 PLN grid resolution, this is the *minimal detectable* shift. The true shift may be smaller.

3. The magnitude of the shift follows clear rules: $\lambda \leq 0.02$ preserves $\text{BDP}_c = 500$; $\lambda \geq 0.05$ shifts it to 750; $\rho \in [0.02, 0.1]$ shifts it to 750; $\rho \geq 0.2$ restores $\text{BDP}_c = 500$.

These findings extend the universality result of Paper IV from the exogenous to the endogenous domain. The phase transition is not an artifact of static parameterization—it reflects a genuine structural feature of the model economy.

5.2 The Superadditivity Effect

The +6.5 pp peak adoption boost in the full endogenous cell (vs. baseline) exceeds what either mechanism achieves alone. This superadditivity arises because the two mechanisms operate on complementary channels: endogenous σ lowers the adoption threshold (making marginal firms more likely to automate), while dynamic rewiring creates preferential hubs that amplify contagion (making adoption spread faster once it starts). The combination produces a stronger positive feedback loop than either channel alone.

From a policy perspective, the superadditivity implies that economies experiencing both rapid learning and network restructuring (e.g., emerging digital ecosystems) will exhibit more pronounced adoption dynamics than models with static technology or networks would predict. The optimal BDP must account for these amplification effects.

5.3 Why Not SOC?

The absence of self-organized criticality is informative. In canonical SOC systems (sandpile models, forest fires), the dynamics *dissipate* energy or material at the boundary, creating a natural mechanism for the system to hover near criticality (Bak 1996). In our model, σ evolution is a *positive* feedback: adoption raises σ, which lowers the threshold, which raises adoption. This is an amplifying loop, not a dissipative one. The ratchet constraint (Equation 2) further prevents the oscillatory dynamics that SOC requires.

For SOC to emerge in this class of models, one would need a *negative* feedback channel—for example, resource depletion, skill obsolescence, or market saturation that reduces σ when adoption is too high. The purely increasing-returns specification used here is insufficient.

5.4 The Non-Monotonic ρ Effect

The non-monotonic dependence of BDP_c on ρ (Figure 10) has an elegant interpretation in terms of network science. At $\rho = 0$, the network is a static Watts–Strogatz small world. At moderate ρ (0.02–0.1), preferential attachment generates scale-free-like hubs that concentrate influence and shift the critical point. But at high ρ (≥ 0.2), the network refreshes so rapidly that any emergent structure is immediately destroyed, and the effective topology reverts to a random graph—which, as Paper IV demonstrated, has the same BDP_c as the small-world network (Maciaszek 2026c).

This mirrors results from the statistical physics of networks, where dynamic rewiring at intermediate rates can produce qualitatively different behavior from both the static and fully-random limits (Albert and Barabási 2002; Castellano et al. 2009). The window $\rho \in [0.02, 0.1]$ defines a "structured rewiring" regime where preferential attachment creates meaningful heterogeneity.

5.5 Limitations

Several limitations should be noted. First, the BDP grid spacing of 250 PLN limits our ability to precisely locate BDP_c. The observed shift from 500 to 750 is the minimal detectable change; a finer grid might reveal a continuous drift rather than a discrete jump. Second, the learning rule (Equation 2) is deliberately simple—a single-parameter logistic growth with ratchet. Richer specifications (e.g., sector-specific λ, learning spillovers across sectors, or depreciation) could produce different dynamics. Third, the dynamic network model replaces bankrupt firms with homogeneous Traditional entrants; heterogeneous entrants (varying technology levels) could alter the results. Fourth, we study only the PLN monetary regime; the EUR regime with SGP constraints may respond differently to endogenization.

5.6 Policy Implications

The key policy message is reassuring: the optimal BDP identified in Papers I–II ($\sim$500 PLN) remains approximately correct even when technology and networks evolve endogenously. Policymakers need not fear that endogenous dynamics will qualitatively alter the cost–benefit calculus of BDP. The quantitative effect is an *increase* in peak adoption of 5–7 pp under endogenous dynamics, suggesting that static models may somewhat *underestimate* the scale of the automation transition, but the optimal intervention point is essentially unchanged.

The sector stratification in σ growth (Figure 4) has a direct policy implication: sector-specific transition support should be concentrated on the high-σ sectors (BPO/SSC, Manufacturing) where endogenous technology dynamics are strongest, rather than spread uniformly. Low-σ sectors (Public, Healthcare) are largely unaffected by endogenous dynamics and do not require additional policy attention beyond what the static model prescribes.

6 Conclusion

This paper tested whether the universality of AI adoption phase transitions—established in Paper IV under static, exogenous conditions—survives when both the technology parameter σ and the inter-firm network are made endogenous. Through a 2×2 factorial

design with 10,080 Monte Carlo simulations, we find that universality is *perturbed but preserved*: the reentrant adoption shape is fully robust across all factorial cells, the critical point shifts by at most 250 PLN, and the effects of the two endogenous mechanisms are superadditive. Endogenous σ does not produce self-organized criticality; instead, a positive feedback loop between adoption and learning amplifies the transition without driving the system to the critical point.

The accumulated evidence across five papers—spanning monetary regimes, empirical calibration, network topology, decision rules, system size, and now endogenous dynamics—establishes that the reentrant phase transition at $\mathrm{BDP}_c \approx 500$ PLN is a fundamental structural feature of the model economy, robust to every dimension of variation we have tested.

References

Albert, Réka and Albert-László Barabási (2002). "Statistical Mechanics of Complex Networks". In: *Reviews of Modern Physics* 74.1, pp. 47–97. DOI: 10.1103/RevModPhys.74.47.

Arrow, Kenneth J. (1962). "The Economic Implications of Learning by Doing". In: *Review of Economic Studies* 29.3, pp. 155–173. DOI: 10.2307/2295952.

Arthur, W. Brian (1989). "Competing Technologies, Increasing Returns, and Lock-In by Historical Events". In: *The Economic Journal* 99.394, pp. 116–131. DOI: 10.2307/2234208.

— (1994). *Increasing Returns and Path Dependence in the Economy*. Ann Arbor: University of Michigan Press.

Bak, Per (1996). *How Nature Works: The Science of Self-Organized Criticality*. New York: Copernicus.

Bak, Per, Chao Tang, and Kurt Wiesenfeld (1987). "Self-Organized Criticality: An Explanation of the $1/f$ Noise". In: *Physical Review Letters* 59.4, pp. 381–384. DOI: 10.1103/PhysRevLett.59.381.

Barabási, Albert-László and Réka Albert (1999). "Emergence of Scaling in Random Networks". In: *Science* 286.5439, pp. 509–512. DOI: 10.1126/science.286.5439.509.

Battiston, Stefano et al. (2012). "Liaisons Dangereuses: Increasing Connectivity, Risk Sharing, and Systemic Risk". In: *Journal of Economic Dynamics and Control* 36.8, pp. 1121–1141. DOI: 10.1016/j.jedc.2012.04.001.

Binder, Kurt and Dieter W. Heermann (2010). *Monte Carlo Simulation in Statistical Physics: An Introduction*. 5th. Berlin: Springer. DOI: 10.1007/978-3-642-03163-2.

Castellano, Claudio, Santo Fortunato, and Vittorio Loreto (2009). "Statistical Physics of Social Dynamics". In: *Reviews of Modern Physics* 81.2, pp. 591–646. DOI: 10.1103/RevModPhys.81.591.

David, Paul A. (1985). "Clio and the Economics of QWERTY". In: *American Economic Review* 75.2, pp. 332–337.

Jackson, Matthew O. and Alison Watts (2002). "The Evolution of Social and Economic Networks". In: *Journal of Economic Theory* 106.2, pp. 265–295. DOI: 10.1006/jeth.2001.2903.

Maciaszek, Mateusz (2026a). "Empirical Estimation of Sectoral Elasticities of Substitution between Labor and AI Capital: A GMM and Bayesian Approach". Working paper.

— (2026b). "Monetary Regime and the Automation Transition: PLN Float versus EUR with Stability and Growth Pact Constraints". Working paper.

Maciaszek, Mateusz (2026c). "Phase Diagram and Universality of AI-Driven Labor Market Transitions: Evidence from a Stock-Flow Consistent Agent-Based Model". Working paper.

— (2026d). "The Acceleration Paradox: How Universal Basic Income Amplifies AI-Driven Labor Market Disruption in an Agent-Based Stock-Flow Consistent Model". Working paper.

Stanley, H. Eugene (1971). *Introduction to Phase Transitions and Critical Phenomena*. New York: Oxford University Press.

Watts, Duncan J. and Steven H. Strogatz (1998). "Collective Dynamics of 'Small-World' Networks". In: *Nature* 393.6684, pp. 440–442. DOI: 10.1038/30918.